

Event Streams of Moving Scenes: Rate Limits and Regeneration Under Affine Motion

[Author list to be set]

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Abstract

An event camera reports the changes of each pixel’s log intensity as asynchronous threshold crossings. When a scene moves across the sensor, every pixel on the path of an edge observes the same intensity profile, so the stream appears to describe that profile many times. It does not repeat exactly: each pixel fires on its own lattice of threshold levels, whose offset, the threshold phase, differs from pixel to pixel, and every event carries timing jitter. We determine what these differences cost a code, for a ramp edge under an affine flow. Losslessly, the phase of each pixel and the jitter of its other events must be described however well the scene and its motion are known, so the rate per event does not fall with the length of the path. Under a mean-squared timing tolerance there are three regimes. Below the jitter variance the code describes every event time. Above it the code describes one number per pixel. Beyond a threshold set by the event spacing of the pixel it describes nothing, because the events can be regenerated from the edge and its motion, and what remains is the motion, at a cost that grows at most logarithmically with the path. The rate-distortion function is exact in the first regime and known within 0.255 bit per pixel in the second. These results lead to a codec that transmits the events of one window as a template and one affine map for each later window, and we quantify what moving events loses against regenerating them. On traffic recordings, a motion-compensated template shortens the lossless code only slightly, the fidelity of regenerated events at a fixed tolerance rises with object speed, and regeneration reaches a given fidelity with a fraction of the events that random thinning needs.

Index Terms: Event camera, send-on-delta sampling, level-crossing sampling, rate-distortion theory, generalized sampling, motion compensation, neuromorphic sensing.

1 Introduction

An event camera has no frames. Each pixel compares its log intensity with a stored reference and emits an event, with a time stamp and a polarity, whenever the difference reaches a contrast threshold [1,2]. The output is a stream

of address-events whose rate follows the scene: it is close to zero for a static scene and rises with the speed and the contrast of what moves. A sensor node that forwards this stream over a rate-limited link must decide how many bits the stream needs.

The stream of a moving scene looks highly redundant. If an object moves across the sensor, the pixels along the path of one of its edges observe the same intensity profile, one after the other. Yet they do not emit the same events. Each pixel compares the intensity with its own reference, so it fires on its own lattice of levels, and the offset of that lattice, the threshold phase of the pixel, differs from pixel to pixel. The same profile therefore produces a different event train at every pixel (Fig. 1). This paper asks what these differences cost: how many bits a code needs to preserve them, and how many it saves when a timing error is allowed.

The setting has precedents in sampling theory. An event pixel is a send-on-delta sampler [3], which for a monotone input coincides with level-crossing sampling [4, 5]. An object that crosses M pixels is sampled by M such samplers, which recalls the generalized sampling expansion [6], by which a bandlimited signal is recovered from the outputs of m linear systems that are each sampled at $1/m$ of the Nyquist rate. In multi-channel time encoding, several integrate-and-fire encoders with shifted integrators share one input, and the shifts are what make the channels complementary [7–9]. The threshold phases play that role here.

Existing event coders approach the rate empirically. Lossless coders exploit the spatial and temporal correlation of addresses and time stamps [10–12]. Lossy coders discard or aggregate events [13, 14], and the coder of [15] suppresses events that an on-sensor optical flow can predict and lets the receiver extrapolate them. A recent survey notes that reported results are hard to compare [16], and no accepted distortion measure for lossy event coding exists [17]. On the theoretical side, rate-distortion results for point processes concern Poisson and finite point processes [18–21]. They do not model events that a moving profile generates through a threshold rule.

This paper models that mechanism and derives what a code can and cannot save. Its central result is the rate-distortion function of the events of a pixel under a timing tolerance. The other results lead to it or follow from it.

1) *Mechanism.* Given the edge and its motion, the events of a pixel are a function of a single number, its threshold phase, up to timing jitter. The phase is a property of the pixel and is independent across pixels. Next to what the pixels of a path share, the edge and its motion, each pixel therefore has something private. A lossless code must describe it for every pixel, and its rate per event does not decrease with the number M of pixels on the path (Theorem 1).

2) *Rate under a timing tolerance.* The tolerance decides which of the private quantities a code must describe (Theorem 2). Below the jitter variance it describes all of them, and the rate is known exactly. Above it one number per pixel remains, and the rate is known within 0.255 bit. Beyond a threshold that grows with the event spacing of the pixel, the contrast threshold divided by the rate of change of its intensity, the rate is zero: the decoder regenerates the events from the edge and its motion. Slowly crossed pixels need a larger tolerance. Along a path, a description of the motion of $O(\ln M)$ nats then replaces a lossless description of order M .

3) *Regeneration in practice.* A sender has the events and not the edge. We replace the edge by the events of a key window, which the receiver moves along a fitted affine map. Moved events keep the phases of the pixels they came from, and we show that this doubles the timing error of the ideal decoder at equal event spacings (Proposition 2). We measure the codec in total bits on traffic recordings [22]. *[To be completed: one sentence on the second dataset [23] once it has been measured.]*

Section 2 states the model. Sections 3 and 4 contain the theory. Section 5 describes the codec, and Section 6 the measurements. Proofs are in the appendices. Rates are in nats unless bits are named, $\|\cdot\|$ is the Euclidean norm of a vector and the spectral norm of a matrix, and $[x]^+ = \max(x, 0)$.

2 Model

We first state the pixel rule, from which the phase mechanism follows for any scene. We then specialize to a ramp edge in uniform passage, for which the rates have closed forms.

2.1 Pixel

Pixels sit at the lattice points $i \in \mathbb{Z}^2$ of the image plane. Pixel i observes the log intensity $\ell_i(t) = L(i, t)$ of a continuous field L .

(A1) Send-on-delta. Pixel i holds a reference μ_i with $|\ell_i - \mu_i| < C$ at the start. It emits an event at the first time t at which $|\ell_i(t) - \mu_i| = C$, with polarity $p = \text{sign}(\ell_i(t) - \mu_i)$, and then sets $\mu_i \leftarrow \mu_i + pC$. The contrast threshold $C > 0$ is common to all pixels and known to the decoder. The *phase* of the pixel is the number $a_i \in (0, C]$ with $\mu_i \in a_i + C\mathbb{Z}$ at the start.

2.2 Scene and Motion

A template $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ gives the log intensity in object coordinates. At time t the object point u is at the image position $\phi_t(u) = J_t u + b_t$, with $J_t \in \mathbb{R}^{2 \times 2}$ invertible, so

$$L(x, t) = g(\phi_t^{-1}(x)) \quad (1)$$

and pixel i observes

$$\ell_i(t) = g(u_i(t)), \quad u_i(t) = J_t^{-1}(i - b_t). \quad (2)$$

The curve u_i is the trajectory of pixel i in object coordinates. The velocity field $v(x, t) = \dot{\phi}_t(\phi_t^{-1}(x))$ is affine in x and satisfies the brightness constancy relation

$$\frac{\partial L}{\partial t}(x, t) + \langle \nabla L(x, t), v(x, t) \rangle = 0, \quad (3)$$

which follows from differentiating $L(\phi_t(u), t) = g(u)$ in t . We call the motion an *affine flow* when the velocity field is also constant in time,

$$v(x) = \Lambda x + v_0, \quad (4)$$

with generator $\Lambda \in \mathbb{R}^{2 \times 2}$. A translation has $\Lambda = 0$. The trace of Λ is the rate at which the object grows or shrinks in area, its antisymmetric part rotates the object, and its symmetric traceless part shears it. At a given instant, any smooth velocity field has the form (4) to first order in x around a point, so (4) is the local model of a general motion over a short time.

2.3 Edge

(A2) Ramp edge. The template is a ramp across a line with unit normal ν ,

$$g(u) = \varrho(\langle \nu, u \rangle - c_0), \quad \varrho(z) = \begin{cases} 0, & z \leq 0, \\ \gamma z, & 0 \leq z \leq w, \\ V, & z \geq w, \end{cases} \quad (5)$$

with slope $\gamma > 0$, width w and height $V = \gamma w = nC$ for an integer $n \geq 1$.

Pixel i sees the edge through its edge coordinate

$$z_i(t) = \langle \nu, u_i(t) \rangle - c_0, \quad \ell_i(t) = \varrho(z_i(t)). \quad (6)$$

(A2') Uniform single passage. The edge passes pixel i once: $z_i < 0$ before a time t_i , $z_i > w$ after the passage, and in between z_i is affine in t with slope $r_i > 0$.

Inside the ramp, $\nabla L = \gamma \nu_t$ with $\nu_t = J_t^{-\top} \nu$ a normal of the edge in the image, and $\dot{z}_i = -\langle \nu_t, v(i, t) \rangle$. By (3), the intensity of pixel i rises during the passage at the rate

$$\gamma r_i = -\langle \nabla L, v \rangle = \|\nabla L\| v_{\perp, i}, \quad (7)$$

with the field evaluated at pixel i . Here $v_{\perp, i} = -\langle \nabla L, v \rangle / \|\nabla L\|$ is the speed at which the edge advances over the pixel along its normal, counted positive when the

intensity of the pixel rises, as it does in a passage with $r_i > 0$. We call

$$\tau_i = \frac{C}{\gamma r_i} = \frac{C}{\|\nabla L\| v_{\perp,i}} \quad (8)$$

the *event spacing* of pixel i . It is the time that the ramp takes to pass the pixel, divided by n . At a fixed image slope $\|\nabla L\|$ it is inversely proportional to the normal speed of the edge. Assumption (A2') holds exactly for a translation, where τ_i is the same for all pixels, and for a plane that approaches the camera along the optical axis at constant speed, where J_t^{-1} is affine in t and b_t is fixed. For an affine flow it holds to first order (Remark 1), and τ_i varies from pixel to pixel.

2.4 Phase, Jitter and Clock

(A3) Phase. Before the edge arrives, pixel i rests at the level 0 with its reference behind the passage, $\mu_i \in (-C, 0]$, so that $a_i = \mu_i + C$. The phases a_i are independent across pixels, uniform on $(0, C]$, and independent of the jitter.

(A4) Jitter and clock. The k th event of pixel i is recorded at $T_{i,k} = t_{i,k} + \varepsilon_{i,k}$, where $t_{i,k}$ is the time given by (A1) and the $\varepsilon_{i,k}$ are independent $\mathcal{N}(0, \sigma^2)$ variables, independent of the phases, with $\sigma > 0$. The stored value is $Y_{i,k} = \delta \lfloor T_{i,k} / \delta \rfloor$ for a clock resolution δ . The index k of each recorded event is known.

The uniform phase does not arise inside the model, in which every pixel rests at the same level. It stands for the offsets of real pixels and for their earlier activity. A pixel that the edge crosses from the high level to the low level is covered by the mirrored assumption, with its reference in $[V, V + C)$. Remark 4 treats a reference that is ahead of the passage.

We write $T_i = (T_{i,1}, \dots, T_{i,n})$ and $Y_i = (Y_{i,1}, \dots, Y_{i,n})$, and $\mathcal{P}$ for the set of pixels that the edge passes during the observation, with $M = |\mathcal{P}|$. Let $\eta(a, \sigma)$ denote the differential entropy of the sum of a uniform variable on $(0, a)$ and an independent $\mathcal{N}(0, \sigma^2)$ variable. It satisfies

$$\frac{1}{2} \ln(a^2 + 2\pi e \sigma^2) \leq \eta(a, \sigma) \leq \frac{1}{2} \ln(2\pi e(a^2/12 + \sigma^2)) \quad (9)$$

by the entropy power inequality [24, Ch. 17] and by the Gaussian bound on the entropy at a given variance [24, Ch. 8], and $\eta(a, \sigma) - \ln a \rightarrow 0$ as $\sigma/a \rightarrow 0$.

3 Structure and Lossless Limit

Lemma 1. *Under (A1), for any continuous ℓ_i , every event of pixel i occurs at a time t with $\ell_i(t) \in a_i + C\mathbb{Z}$.*

The lattice $a_i + C\mathbb{Z}$ never changes. The phase is a property of the pixel, set by its history and not by the edge that passes.

Lemma 2. *Under (A1), (A2), (A2') and $\mu_i \in (-C, 0]$ before the passage, pixel i emits exactly n events, all of*

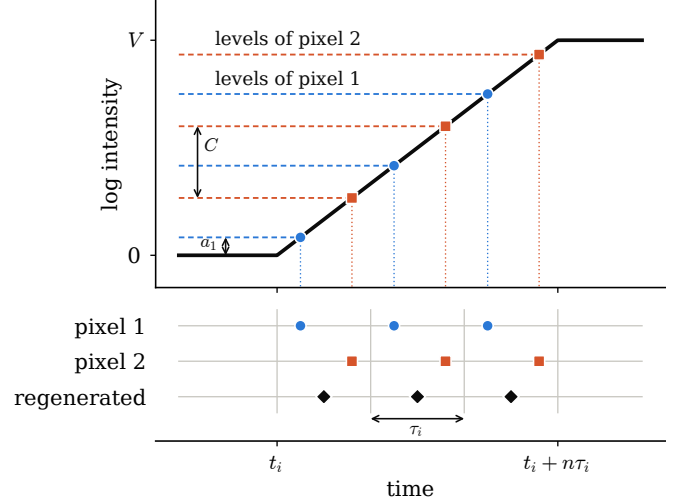


Figure 1: One ramp, two pixels with different threshold phases, and the regenerated events (22). Time is measured from the arrival of the edge at each pixel. Jitter is omitted.

positive polarity, at

$$t_{i,k} = t_i + A_i + (k-1)\tau_i, \quad k = 1, \dots, n, \quad (10)$$

where $A_i = a_i/(\gamma r_i)$. Under (A3), A_i is uniform on $(0, \tau_i]$.

Given the template and the flow, the noise-free events of a pixel are thus a function of one number, its phase. Fig. 1 shows two pixels with $n = 3$ that see the same ramp: the intensity trajectories coincide, and the event trains differ by the difference of the phases. In the object coordinate, an event is a point of the graph of the profile on the level lattice of its pixel: the M pixels of a path sample the same profile on M lattices with independent offsets.

Theorem 1 (Lossless code length). *Under (A1) to (A4) and (A2'), the vectors T_i , $i \in \mathcal{P}$, are independent given the template and the flow,*

$$h(T_i) = \frac{n-1}{2} \ln(2\pi e \sigma^2) + \eta(\sqrt{n} \tau_i, \sigma), \quad (11)$$

and

$$H(Y_i | g, \phi) = n \ln \frac{1}{\delta} + h(T_i) + e_\delta, \quad (12)$$

where $e_\delta \geq 0$ for every δ , and $e_\delta \rightarrow 0$ as $\delta/\sigma \rightarrow 0$, uniformly in τ_i .

When $\sigma \ll \tau_i$ and $\delta \ll \sigma$, (12) gives the cost per event

$$\frac{H(Y_i | g, \phi)}{n} \approx \frac{n-1}{n} \ln \frac{\sigma \sqrt{2\pi e}}{\delta} + \frac{1}{n} \ln \frac{\sqrt{n} \tau_i}{\delta}. \quad (13)$$

One event of each pixel pays for the phase and the other $n-1$ pay for jitter. The code length of the path is the sum of (12) over $\mathcal{P}$. It grows in proportion to M , and neither

term of (13) falls as the path gets longer: a decoder that knows the edge and its motion exactly still pays (13) for every event. What the pixels of a path share, the edge and its motion, a code can describe once. What is private to each pixel, its phase and its jitter, it must describe for each pixel, and (12) is that irreducible conditional rate. Section 6 measures how much more a code pays that does not know the motion.

Remark 1 (Accuracy of the uniform passage). For an affine flow (4), $\dot{\nu}_t = -\Lambda^\top \nu_t$ and $\dot{z}_i = \langle \nu_t, \Lambda v(i) \rangle$, hence $|\dot{z}_i| \leq \kappa_i |\dot{z}_i|$ with $\kappa_i = \|\Lambda\|/|\cos \vartheta_i|$, where ϑ_i is the angle between the velocity $v(i)$ and the normal ν_t . Take r_i as the slope of z_i at t_i . Over a passage of duration $P_i = n\tau_i$, the event times then deviate from (10) by at most $\kappa_i P_i^2/2$, to first order in $\kappa_i P_i$. For a generator of norm 0.3 s^{-1} , a passage of 5 ms and motion along the normal this is $4\text{ }\mu\text{s}$, three orders of magnitude below the passage time. The bound grows without limit for edges that are nearly parallel to their motion. Such edges pass few pixels per unit time and contribute few events.

Remark 2 (Several edges). Theorem 1 counts one edge that passes each pixel once. If further edges of the same object pass the same pixel, they meet the same lattice, so the phase is paid once per pixel and not once per edge. With many edges per pixel the cost per event tends to the jitter term $\ln(\sigma\sqrt{2\pi e}/\delta)$, which does not depend on the path either.

4 Rate Under a Timing Tolerance

The encoder and the decoder know the template and the flow. The decoder outputs $\hat{T}_i \in \mathbb{R}^n$ for each pixel, and the distortion is the mean-squared timing error

$$d(T_i, \hat{T}_i) = \frac{1}{n} \|T_i - \hat{T}_i\|^2. \quad (14)$$

4.1 One Pixel

Let $R_i(D) = \inf I(T_i; \hat{T}_i)$, with the infimum over the conditional laws of $\hat{T}_i$ that satisfy $\mathbb{E} d(T_i, \hat{T}_i) \leq D$, be the rate-distortion function of the events of pixel i . It is the rate that block codes over many independent passages approach. Define

$$s_i^2 = \frac{n\tau_i^2}{12} + \sigma^2, \quad D_{A,i} = \sigma^2 + \frac{\tau_i^2}{12}, \quad (15)$$

$$\Gamma_i = \frac{1}{2} \ln(2\pi e s_i^2) - \eta(\sqrt{n}\tau_i, \sigma). \quad (16)$$

By (9), $0 \leq \Gamma_i \leq \frac{1}{2} \ln(2\pi e/12)$, which is 0.255 bit, for every σ and τ_i .

Theorem 2 (Rate-distortion function of a pixel). *Under (A1) to (A4) and (A2'),*

$$R_i(D) = h(T_i) - \frac{n}{2} \ln(2\pi e D) \quad \text{for } 0 < D \leq \sigma^2, \quad (17)$$

$$[U_i(D) - \Gamma_i]^+ \leq R_i(D) \leq U_i(D) \quad \text{for } \sigma^2 < D < D_{A,i}, \quad (18)$$

$$R_i(D) = 0 \quad \text{for } D \geq D_{A,i}, \quad (19)$$

with

$$U_i(D) = \frac{1}{2} \ln \frac{s_i^2}{nD - (n-1)\sigma^2}. \quad (20)$$

Moreover,

$$R_i(D) \geq \frac{n(D_{A,i} - D)}{2s_i^2} \quad \text{for } 0 < D \leq D_{A,i}, \quad (21)$$

so that $R_i(D) > 0$ for $D < D_{A,i}$.

The three regimes have a direct reading. Below the jitter variance, every bit of timing precision costs a bit, per event: the rate falls by $\log_2 10 = 3.32$ bits per event and per decade of the root-mean-square tolerance $\sqrt{D}$. Between σ^2 and $D_{A,i}$, the $n-1$ jitter components orthogonal to a common shift of the events are no longer described, and the code describes one number per pixel, the phase together with the mean jitter, to within 0.255 bit. From $D_{A,i}$ on, nothing is described. The decoder that attains (19) outputs the mean times

$$\hat{T}_{i,k} = t_i + (k - \frac{1}{2})\tau_i, \quad k = 1, \dots, n. \quad (22)$$

It *regenerates* the events of the pixel from the edge and the flow. Each regenerated event is within $\tau_i/2$ of its noise-free time, and the count and the polarities are exact.

Two of these statements extend beyond the ramp. If the noise-free event times of a pixel are any function of its phase, (17) holds below the jitter variance with the entropy of the actual event times, and the smallest distortion at zero rate is $\frac{1}{n} \text{tr Cov}(T_i)$, attained by the mean times. The middle regime, in which one number per pixel is described to within 0.255 bit, and the closed forms (11), (20) and (15) belong to the ramp in uniform passage, where the phase shifts all events of a pixel by a common time.

The threshold $D_{A,i}$ grows with the square of the event spacing. By (8), a pixel whose intensity changes slowly has a large spacing and needs a large tolerance before its events can be regenerated, and a pixel that is crossed quickly needs a small one.

4.2 A Path

For the M pixels of a path, let the distortion be the average of (14) over $i \in \mathcal{P}$, and let $R_{\mathcal{P}}(D)$ be the rate-distortion function of $(T_i)_{i \in \mathcal{P}}$ under it.

Theorem 3 (Rate-distortion function of a path). *Under (A1) to (A4) and (A2'),*

$$R_{\mathcal{P}}(D) = \min \left\{ \sum_{i \in \mathcal{P}} R_i(D_i) : \frac{1}{M} \sum_{i \in \mathcal{P}} D_i \leq D \right\}. \quad (23)$$

In particular,

$$R_{\mathcal{P}}(D) = \sum_{i \in \mathcal{P}} h(T_i) - \frac{Mn}{2} \ln(2\pi e D) \quad \text{for } 0 < D \leq \sigma^2, \quad (24)$$

and $R_{\mathcal{P}}(D) = 0$ if and only if

$$D \geq \bar{D}_A = \sigma^2 + \frac{1}{12M} \sum_{i \in \mathcal{P}} \tau_i^2. \quad (25)$$

If $\tau_i \leq \tau_{\max}$ on the path, then for $D \leq \bar{D}_A$

$$R_{\mathcal{P}}(D) \geq \frac{nM(\bar{D}_A - D)}{2(n\tau_{\max}^2/12 + \sigma^2)}. \quad (26)$$

If the tolerance is imposed on every pixel separately, the rate is zero if and only if $D \geq \max_i D_{A,i}$, and the slowest crossing sets the tolerance. By (26), below $\bar{D}_A$ a rate that is proportional to M remains.

4.3 The Motion

Theorems 2 and 3 assume that the decoder knows the flow. It remains to count what describing the flow costs.

Proposition 1 (Description of the motion). *Let the flow depend on a parameter $\theta = (\theta_1, \dots, \theta_q)$ that ranges over a box whose j th side has length at most $L_j M^{\alpha_j}$, and let the mean times $\bar{t}_{i,k}(\theta) = t_i(\theta) + (k - \frac{1}{2})\tau_i(\theta)$ of (22) satisfy $|\partial \bar{t}_{i,k} / \partial \theta_j| \leq S_j M^{\beta_j}$ on the box, for all events of the path, with constants $L_j, S_j > 0$ and exponents $\alpha_j, \beta_j \geq 0$ that do not depend on M . Then for every $\xi > 0$ the events of the path can be described with average distortion at most $(1 + \xi)\bar{D}_A$ using*

$$B \leq \left(\sum_{j=1}^q (\alpha_j + \beta_j) \right) \ln M + O(1) \quad \text{nats}, \quad (27)$$

where the $O(1)$ term depends on q, ξ, σ and the constants and not on M .

For a translation along a line with a speed in a fixed interval that excludes zero, θ holds the position of the edge, with a range of order M and a sensitivity of order one, and the speed, with a range of order one and a sensitivity of order M , so $B \leq 2 \ln M + O(1)$. The same holds with finite exponents, and hence $B = O(\ln M)$, for an affine flow whose normal speed is bounded away from zero on the path and whose duration is polynomial in M . The lossless cost of the same events is the sum of (12), of order M , so the ratio of the two grows at least as $M / \ln M$. Above $\bar{D}_A$ the description that remains is $O(\ln M)$. Below it, a rate proportional to M remains. The bound (27) is for a receiver that holds the edge. Section 5 accounts for the template and the regions.

Remark 3 (Event count under a change of scale). An edge segment emits n events for every pixel that it passes, so its event rate is n times the area that it sweeps per unit time,

that is n times its length times its mean normal speed. Under a similarity flow about a fixed point, growth by a factor ζ makes the segment ζ times longer and ζ times faster, and the rate grows as ζ^2 . For an object that approaches at constant speed, the image speed itself grows as ζ^2 and the rate as ζ^3 . A regeneration that moves a fixed set of events keeps their number. Under a change of scale it must also change the number of events, or regenerate them from the edge.

Remark 4 (Departures from the model). *Reference ahead of the passage.* If $\mu_i \in (0, C)$ before a rising passage, as after a passage of the opposite polarity at the same pixel, the first level that the intensity meets is the reference itself, which does not fire. The pixel then emits $n - 1$ events, and the results hold with $n - 1$ in place of n and $t_i + \tau_i$ in place of t_i . *Height.* If the height of the edge is not a multiple of the threshold, $V/C = n + f$ with $0 < f < 1$, pixel i emits $n + 1$ events when $a_i \leq fC$ and n otherwise. Reproducing the count exactly costs the binary entropy of f per pixel, and a zero-rate statement then holds only for a distortion measure that tolerates one unmatched event per pixel, with a modified threshold. *Sensor.* Noise events, threshold mismatch between pixels, a refractory time and a jitter that depends on the slope are outside the model, and their effect on the three regimes is not analyzed here.

5 A Regenerating Codec

The decoder (22) needs the edge and the flow. A practical sender has neither. It has the events. The codec of this section replaces the edge by events that the receiver already holds and the flow by a map that the sender fits.

5.1 Template, Motion and Groups

Time is cut into windows of length T_0 , and the windows of a region form groups of K . In the first window of a group, the key window, the sender transmits the events of the region. For each of the following $K - 1$ windows it transmits one affine map. The receiver moves every key event (x, p, t) to the pixel nearest to the image of x under the map and to the time $t + mT_0$, where m counts the windows since the key window, and it keeps the polarity. The moved events are its reconstruction of window m . Under an affine flow, the map from one instant to an instant mT_0 later is the same for all instants of a window, so one map per window is exact for the positions of the edges. It does not change the number of events, which a change of scale does (Remark 3).

An affine map is described by six integers: a displacement in pixels, and four shape parameters that give the displacement of the edges of the bounding rectangle of the key events, two for a stretch along each axis and two for shear and rotation. With all shape parameters zero the map is a translation.

The sender chooses the map of each window to maximize the fidelity (29) of the moved key events against the true events of that window on one fixed working grid. Results on other grids use the same maps. The search is coarse to fine, starts from the displacement predicted by the velocity of the region, and never returns a map that is worse than the translation.

5.2 Moved Events Against Regenerated Events

The codec differs from the decoder (22) in three ways. It moves the phases of the key pixels where (22) outputs the mean over the phase, it moves the jitter of the key events, and it moves a fixed number of events. The first two have a price that the model gives exactly.

Proposition 2 (Moved events). *Let the same edge pass pixel j in the key window and a different pixel i later, under (A1) to (A4) and (A2'), and let the map be exact, so that the k th event of pixel j is moved to pixel i at the time $\tilde{T}_{i,k} = T_{j,k} + t_i - t_j$. Then*

$$\frac{1}{n} \mathbb{E} \|T_i - \tilde{T}_i\|^2 = D_{A,i} + D_{A,j} + \frac{4n^2 - 1}{12} (\tau_i - \tau_j)^2. \quad (28)$$

With equal event spacings the error is $2D_A$, twice that of (22): the codec trades a factor of two in the timing error for not needing the edge. Unequal spacings, as under a change of scale, add the last term, which grows with the square of the number of events per pixel. If the moved events fall on their own pixel, as for an object that is nearly at rest, the phases coincide and only the jitter term $2\sigma^2$ remains. The third difference, the number of events, is not a timing error. It is measured in Section 6.

5.3 Receiver Representation and Fidelity

The theory uses the timing error (14), which presupposes that regenerated and true events are paired. For measured streams we use the representation that event-based receivers compute, a voxel grid: the number of events of each polarity in blocks of $\Delta_x \times \Delta_x$ pixels and in time bins of length Δ_t . For a regenerated set with voxel counts c' and a true set with counts c , of sizes N' and N , the fidelity is

$$F = \frac{2 \sum \min(c', c)}{N' + N}, \quad (29)$$

the share of the events of the two sets that can be paired one to one inside a voxel. A subset that holds a share χ of the events has $F = 2\chi/(1 + \chi)$ on every grid, whichever events it keeps. For a codec that sends a share χ of the events of a group and reaches the fidelity F over the group, the key window included, we report

$$G = \frac{F/(2 - F)}{\chi}, \quad (30)$$

the factor by which it lowers the number of events sent at equal fidelity against thinning. The factor counts events

and serves as an explanation. The main results are in total bits, which include the key events, the maps and the description of the regions.

The fidelity (29) and the timing error (14) are different distortion measures, and the bin length plays the role of the tolerance. One case connects them. For blocks of one pixel, a train that is long against the event spacing, and a jitter that is small against the bin, Lemma 2 gives the fidelity of the regenerated events (22) directly: a bin holds at most one event of each train when $\Delta_t \leq \tau_i$, the offset between the two trains is uniform, and in expectation over the phase

$$F = \frac{\Delta_t}{\tau_i} \quad \text{for } \Delta_t \leq \tau_i, \quad (31)$$

with $F = 1$ at $\Delta_t = \tau_i$. The voxel counts of a pixel are thus reproduced only when the bin is at least as long as its event spacing, the same scaling as the threshold of Theorem 2, for which $\sqrt{D_{A,i}} = \tau_i/\sqrt{12}$ when $\sigma \ll \tau_i$. In larger blocks the independent phases of the pixels average, and the fidelity is higher at a given bin length.

[To be completed: Regions. In the measurements so far a region is an annotated object. The paper needs regions that the sender finds without annotation.]

6 Experiments

[To be completed: This section is an outline. It is organized by four questions, one figure each. The protocol below is fixed before the simulations are run. Numbers appear only in this section.]

6.1 Protocol

[To be completed: (i) Controlled experiments use synthetic pixels under the model and the timing error (14). Recorded experiments use the voxel fidelity (29) and total bits. (ii) One fixed working grid for fitting the maps. Other grids are evaluated with the same maps. (iii) Baselines: thinning, whose fidelity is $2\chi/(1 + \chi)$ by construction; repeating the key events without motion; the translation alone; and coding all events after quantizing them to the receiver grid. (iv) Regions: annotated objects as the ideal case, and one automatic scheme, with the bits that describe the regions counted. (v) Main axis: total bits against fidelity.]

6.2 The Three Regimes

Fig. 2 compares Theorem 2 with a rate-distortion function that is computed without it, for one pixel with $n = 3$ and $\sigma/\tau_i = 0.05$. These values illustrate the theorem and are not those of a particular sensor. By (32), the event times are an orthogonal image of the scalar X_i , which carries the phase, and of $n - 1$ normal variables. The iteration of [25] gives the rate-distortion function of X_i on a grid of 1217 points, and the normal components enter in closed form at the same slope.

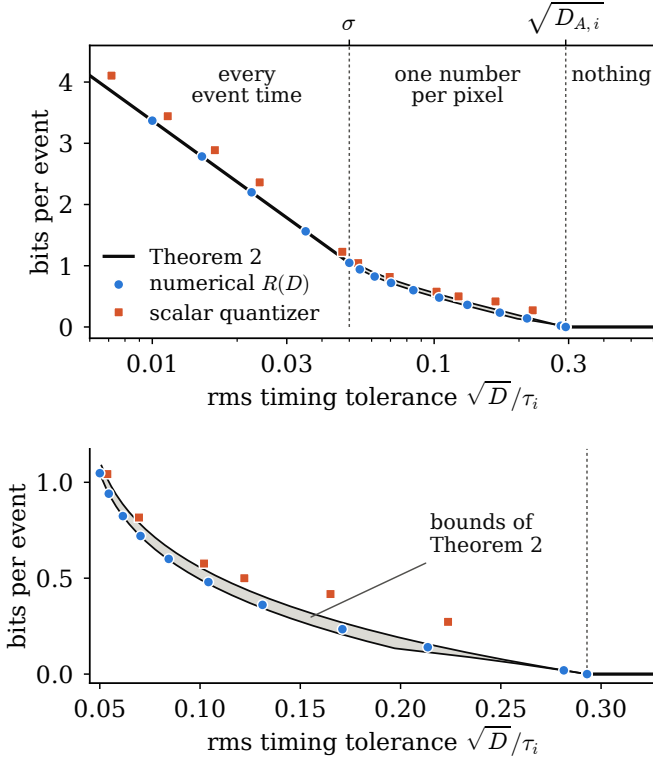


Figure 2: Rate of one pixel against the timing tolerance for $n = 3$ and $\sigma/\tau_i = 0.05$: Theorem 2, a numerical rate-distortion function and a scalar quantizer. Bottom: the regime between σ and $\sqrt{D_{A,i}}$.

Below σ^2 the numerical rate equals (17) to within 4×10^{-6} nat. Between σ^2 and $D_{A,i}$ it stays inside the band between the bounds (18) and (21), which is at most $\Gamma_i = 0.187$ bit per pixel wide here, against the limit of 0.255 bit. The rate leaves σ^2 on the lower bound $U_i - \Gamma_i$: the phase is still described finely there, and the code profits fully from its distribution, which is uniform and not normal. Toward $D_{A,i}$ the description becomes coarse, the rate depends on the phase only through its variance, and the numerical rate, the bound (21) and U_i meet. From $D_{A,i}$ on the rate is zero, and the regenerated events (22) of 2×10^6 simulated pixels have the distortion $0.9994 D_{A,i}$.

The squares are a uniform scalar quantizer with entropy coding, applied to what each regime describes. On every component of the event times it stays 0.254 to 0.258 bit per event above (17). This is the loss $\frac{1}{2} \ln(2\pi e/12)$ of a uniform quantizer at fine resolution, the difference between the entropy $h - \ln \Delta$ of the index of a quantizer of step Δ and (17) at the distortion $\Delta^2/12$. On the mean event time alone, with the other components left out, it stays 0.09 to 0.19 bit per event above the lower bound up to $D_{A,i}$. A code that describes every event time, then one number per pixel, then nothing, is therefore within 0.26 bit per event of the limit in all three regimes.

6.3 Total Bits Against Fidelity

[To be completed: Fig. 3, recorded [22]. The codec for groups of K windows against the baselines of the protocol, in total bits. The lossless code length, with and without a motion-compensated template, is a separate reference: a fidelity of one on a voxel grid is not a lossless reconstruction. The second dataset [23] and one downstream task go into one table. Sources so far: directives 001 to 004.]

6.4 Where Regeneration Stops Working

[To be completed: Fig. 4, recorded, two panels. Is the tolerance large enough: fidelity against bin length, by object speed. Equation (31) is shown only on controlled data and as the ideal case of mean-time regeneration, not as a prediction for the codec, which moves events. Speed alone does not fix τ_i , which also depends on the gradient, the threshold and the direction of motion; an empirical event-rate model that is linear in speed is reported in [26]. Is the template still valid: fidelity against the distance covered since the key window, for the translation and the affine map. Legends separate the ideal regeneration (22), available on controlled data only, from the codec and from the baselines. The event count against the growth of the object (Remark 3) and the controlled test of (28) go to an appendix. Sources so far: directives 002 to 005.]

7 Conclusion

[To be completed: To be written when Section 6 is complete.]

A Proofs of Lemmas 1 and 2

Proof of Lemma 1. The reference starts in $a_i + C\mathbb{Z}$ by the definition of the phase. By (A1) it changes by $\pm C$, so it stays in this set, and an event requires $\ell_i = \mu_i \pm C$. $\square$

Proof of Lemma 2. By (A2) and (A2'), ℓ_i is nondecreasing, equal to 0 before t_i , to $\gamma r_i(t - t_i)$ during the passage, and to V after it. Since $\mu_i \leq 0 = \ell_i$ before the edge arrives and $\mu_i = a_i - C$, the events occur when ℓ_i reaches the levels $a_i + (k-1)C$, $k = 1, 2, \dots$, as long as the level is at most $V = nC$. Because $0 < a_i \leq C$, exactly the levels $k = 1, \dots, n$ are reached. Solving $\gamma r_i(t - t_i) = a_i + (k-1)C$ gives (10), and a_i uniform on $(0, C]$ makes A_i uniform on $(0, \tau_i]$. $\square$

B Proof of Theorem 1

By Lemma 2, $T_i = c_i + A_i \mathbf{1} + \varepsilon_i$ with c_i a function of the template and the flow, $\mathbf{1}$ the all-ones vector, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_n)$. Independence over i follows from (A3) and

(A4). Let Q be an orthogonal matrix whose first row is $\mathbf{1}^\top/\sqrt{n}$. Then $Q(T_i - c_i) = (X_i, W_i)$ with

$$X_i = \sqrt{n} A_i + \varepsilon_i^\parallel, \quad W_i \sim \mathcal{N}(0, \sigma^2 I_{n-1}), \quad (32)$$

where $\varepsilon_i^\parallel \sim \mathcal{N}(0, \sigma^2)$. The variables X_i and W_i are independent, because $Q\varepsilon_i$ has independent coordinates and A_i is independent of ε_i . Differential entropy is invariant under translations and orthogonal maps, so $h(T_i) = h(X_i) + h(W_i)$, which is (11).

Let f be the density of T_i and f_δ its average over the cells of the clock lattice. A cell has probability $\delta^n f_\delta$, so $H(Y_i | g, \phi) = n \ln(1/\delta) - \int f \ln f_\delta$ and $e_\delta = \int f \ln(f/f_\delta)$ is a relative entropy, which is nonnegative. The density f is a mixture over A_i of $\mathcal{N}(m, \sigma^2 I_n)$ densities and f_δ is the same mixture of their cell averages. By the joint convexity of relative entropy, e_δ is at most its largest value for a single $\mathcal{N}(m, \sigma^2 I_n)$ density, which is n times the value for a scalar normal density and depends only on δ/σ and on the offset of m in a cell. It tends to zero with δ/σ , uniformly in the offset [24, Ch. 8]. ■

C Proof of Theorem 2

The distortion (14) is invariant under the translation by c_i and the orthogonal map Q , so R_i is the rate-distortion function of the independent components (X_i, W_i) of (32) under the total squared error nD . Their variances are s_i^2 and σ^2 , the latter $n-1$ times. We drop the index i and write η for $\eta(\sqrt{n}\tau_i, \sigma)$ and $\ln^+ x = [\ln x]^+$.

Lower bound (17) and (18). Let $(\hat{X}, \hat{W})$ have component errors d_X and $d_1, \dots, d_{n-1}$ with $d_X + \sum_j d_j \leq nD$. Independence of the components gives $I(X, W; \hat{X}, \hat{W}) \geq I(X; \hat{X}) + \sum_j I(W_j; \hat{W}_j)$. For a Gaussian component, $I(W_j; \hat{W}_j) \geq \frac{1}{2} \ln^+(\sigma^2/d_j)$, and $I(X; \hat{X}) \geq h(X) - h(X - \hat{X}) \geq \eta - \frac{1}{2} \ln(2\pi e d_X)$. The function $\ln^+(\sigma^2/d)$ is convex in d , so the Gaussian terms are smallest when all d_j equal a common d_W , and $d_W > \sigma^2$ brings no benefit. For $n \geq 2$ this gives

$$R(D) \geq \min_{d_W} \left[\eta - \frac{1}{2} \ln(2\pi e (nD - (n-1)d_W)) + \frac{n-1}{2} \ln \frac{\sigma^2}{d_W} \right], \quad (33)$$

with the minimum over $0 < d_W \leq \sigma^2$, $d_W < nD/(n-1)$. The bracket is convex in d_W and its derivative vanishes only at $d_W = D$. The minimum is at $d_W = D$ when $D \leq \sigma^2$, where it equals the right side of (17), and at $d_W = \sigma^2$ otherwise, where it equals $U(D) - \Gamma$. For $n = 1$ there is no W and the same values follow with $d_X = D$.

Equality for $D \leq \sigma^2$. Write $\varepsilon = \varepsilon' + Z$ with independent $\varepsilon' \sim \mathcal{N}(0, (\sigma^2 - D)I_n)$ and $Z \sim \mathcal{N}(0, DI_n)$, and let $\hat{T} = c + A\mathbf{1} + \varepsilon'$. Then $T = \hat{T} + Z$ with Z independent of $\hat{T}$, the distortion is D , and $I(T; \hat{T}) = h(T) - h(Z)$, which is the lower bound.

Upper bound for $\sigma^2 < D < D_A$. Spend no rate on W , at error σ^2 per component, and describe X at error $d = nD - (n-1)\sigma^2 \in (\sigma^2, s^2)$ through $\hat{X} = (1 - d/s^2)(X - \mathbb{E}X + Z') + \mathbb{E}X$, with Z' an independent normal variable of variance $s^2 d/(s^2 - d)$. The error is d , and $I(X; \hat{X}) = I(X; X + Z') \leq \frac{1}{2} \ln(s^2/d)$, because a normal input maximizes the mutual information across an additive normal noise channel at a given input variance [24].

Lower bound (21) and zero rate. For a variable Ξ with $\ln \mathbb{E} e^{\lambda(\Xi - \mathbb{E}\Xi)} \leq \lambda^2 \varsigma^2/2$ for all λ under a law P , and any law Q , the nonnegativity of the relative entropy between Q and the law with density proportional to $e^{\lambda\Xi}$ with respect to P gives $D(Q\|P) \geq \lambda(\mathbb{E}_Q \Xi - \mathbb{E}_P \Xi) - \lambda^2 \varsigma^2/2$, and the best λ gives $(\mathbb{E}_Q \Xi - \mathbb{E}_P \Xi)^2 \leq 2\varsigma^2 D(Q\|P)$. With P the law of X and Q its conditional law given $\hat{T}$, averaging over $\hat{T}$ yields $\text{Var } \mathbb{E}[X | \hat{T}] \leq 2\varsigma^2 I(X; \hat{T})$. The uniform part of X has width $a = \sqrt{n}\tau$ and satisfies the condition with $a^2/12$, since $\ln(\sinh x/x) \leq x^2/6$, so X satisfies it with $\varsigma^2 = s^2$, and each W_j with $\sigma^2 \leq s^2$. Since the conditional mean minimizes the squared error and the components are independent,

$$\begin{aligned} nD &\geq \mathbb{E}\|T - \mathbb{E}[T | \hat{T}]\|^2 \\ &= nD_A - \text{Var } \mathbb{E}[X | \hat{T}] - \sum_j \text{Var } \mathbb{E}[W_j | \hat{T}] \\ &\geq nD_A - 2s^2 I(T; \hat{T}), \end{aligned} \quad (34)$$

which is (21). With no description the best output is the mean, which is (22), at distortion $\frac{1}{n}(s^2 + (n-1)\sigma^2) = D_A$, so (19) holds. ■

D Proof of Theorem 3

The vectors T_i are independent and the distortion is additive over i . For any $\hat{T}$ with average distortion at most D , let $D_i = \mathbb{E}d(T_i, \hat{T}_i)$. Then $I((T_i)_i; (\hat{T}_i)_i) \geq \sum_i I(T_i; \hat{T}_i) \geq \sum_i R_i(D_i)$ with $\frac{1}{M} \sum_i D_i \leq D$. Conversely, independent test channels whose rates are within any given margin of $R_i(D_i)$ come within M times that margin of the sum. Each R_i is convex, nonincreasing and continuous, and tends to infinity as $D_i \rightarrow 0$ by (17), so the minimum in (23) exists.

For $D < \sigma^2$, every R_i is differentiable at D with the derivative $-n/(2D)$ by (17), and it is convex on $(0, \infty)$. Hence $R_i(D_i) \geq R_i(D) - \frac{n}{2D}(D_i - D)$ for every $D_i > 0$, and summing over i with $\sum_i (D_i - D) \leq 0$ shows that the allocation $D_i = D$ is optimal. This gives (24) for $D < \sigma^2$, and the case $D = \sigma^2$ follows by continuity.

By Theorem 2, $\sum_i R_i(D_i) = 0$ if and only if $D_i \geq D_{A,i}$ for every i , and such an allocation with average at most D exists if and only if $D \geq \frac{1}{M} \sum_i D_{A,i} = \bar{D}_A$. Finally, (21) and $s_i^2 \leq n\tau_{\max}^2/12 + \sigma^2$ give $\sum_i R_i(D_i) \geq n \sum_i [D_{A,i} - D_i]^+ / (2(n\tau_{\max}^2/12 + \sigma^2))$, and $\sum_i [D_{A,i} - D_i]^+ \geq M(\bar{D}_A - D)$, which is (26). ■

E Proof of Proposition 1

Divide side j of the box into the smallest number of equal cells of length at most $\varsigma_j = \sqrt{\xi} \sigma / (q S_j M^{\beta_j})$, and let $\hat{\theta}$ be the center of the cell that contains θ . Side j has at most $L_j M^{\alpha_j} / \varsigma_j + 1$ cells, so the index of $\hat{\theta}$ takes $\sum_j \ln(L_j S_j q M^{\alpha_j + \beta_j} / (\sqrt{\xi} \sigma) + 1)$ nats, which is (27). The decoder outputs $\hat{T}_{i,k} = \bar{t}_{i,k}(\hat{\theta})$. The segment from θ to $\hat{\theta}$ lies in the box, so by the mean value theorem $e_{i,k} = \bar{t}_{i,k}(\theta) - \bar{t}_{i,k}(\hat{\theta})$ satisfies $|e_{i,k}| \leq \sum_j S_j M^{\beta_j} \varsigma_j / 2 \leq \sqrt{\xi} \sigma / 2$. By (10), $T_{i,k} - \bar{t}_{i,k}(\theta) = A_i - \tau_i / 2 + \varepsilon_{i,k}$ has mean zero given θ and variance $D_{A,i}$, and $e_{i,k}$ is a function of θ . Hence $\mathbb{E}(T_{i,k} - \hat{T}_{i,k})^2 = D_{A,i} + \mathbb{E} e_{i,k}^2 \leq D_{A,i} + \xi \sigma^2 / 4$, and the average over the path is at most $\bar{D}_A + \xi \sigma^2 / 4 \leq (1 + \xi) \bar{D}_A$. ■

F Proof of Proposition 2

By (10) and (A4), $T_{i,k} - \bar{T}_{i,k} = (A_i - A_j) + (k - 1)(\tau_i - \tau_j) + \varepsilon_{i,k} - \varepsilon_{j,k}$. The phases and the jitter of the two pixels are independent, so this difference has the mean $(k - \frac{1}{2})(\tau_i - \tau_j)$ and the variance $\tau_i^2 / 12 + \tau_j^2 / 12 + 2\sigma^2 = D_{A,i} + D_{A,j}$. Its mean square is the variance plus the squared mean, and $\frac{1}{n} \sum_{k=1}^n (k - \frac{1}{2})^2 = (4n^2 - 1) / 12$. ■

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